

DNSid — a durable ownership anchor for AI agents

draft-ihsanullah-dnsid-01 · datatracker.ietf.org/doc/draft-ihsanullah-dnsid

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The question it answers: who is accountable for an agent — and does that binding survive key rotation, revocation, and transfer?

Three questions, three anchors

- **Who is accountable?** → a registrant domain + a signed DNS record
 - a registrable domain has a registrant, a contract, and a dispute process behind it
- **Is it really them, right now?** → keys
 - proves control of a key at this moment; says nothing about who stands behind it
- **What has happened over time?** → an append-only lifecycle log
 - DNS proves current control, not history

Ownership is a **fact** — binary, no judgement attached.

“Should I trust this agent” is a **judgement** — policy-dependent, explicitly out of scope.

The record

- One TXT record at `_dnsid.<agent-fqdn>` — the `_dmarc` / `_domainkey` pattern (RFC 8552)
- Ten tags: seven required, three optional
 - `gi` governance identifier — the accountable entity
 - `ek` entity record-signing keys · `ku` agent operational keys (JWKS over HTTPS)
 - `lr` lifecycle log reference · `su` live status · `sg` entity signature
- **Pointer set, not a data store** — no inline keys, no service endpoint, no SVCB
 - nothing here competes with discovery
- `sg` is signed by the **entity** key at `ek`, never by the agent key at `ku`
 - the agent cannot sign its own accountability

Lifecycle

Six states, forward-only

PENDING → PROVISIONING → VERIFYING → ACTIVE → { RETIRED | REVOKED }

- Neither terminal state returns to ACTIVE — re-register, don't resurrect
 - **RETIRED**: graceful end; work signed before it is not invalidated
 - **REVOKED**: forced end, reason code required; work from the triggering period is suspect
- Logged events: ISSUANCE, KEY_ROTATION, REVOCATION, RETIREMENT, MIGRATION (+ optional DELEGATION)
 - ISSUANCE is **bilateral** — entity-signed, agent-countersigned; neither binds the other unilaterally
 - KEY_ROTATION is signed by the *previous operational key* — routine rotation keeps the accountable entity out of the loop
- Log interface is abstract; a SCITT transparency-service entry is the worked example (`lr=scitt:...`)

Applying the taxonomy

§5.1 method — profile = the root-to-leaf paths that are substantive requirements; omit the incidental

- **Identity => Agent Naming and Addressing** — [primary focus; gap]
 - FQDN as stable identifier, decoupled from any locator (§4.4.1's identifier/locator split); your Figure 2 already marks this leaf (gap)
- **Security and Trust => Accountability** — (gap)
 - §4.2.3: audit evidence is only as strong as the identity evidence it references — this use case is that identity leg
- **Operations and Management => Lifecycle Management** — (gap)
 - the *identity-lifecycle* subset: registration, revocation vs retirement, transfer — which §4.7.3 doesn't currently list; not deployment, scaling, or drain
- **Identity => Credential Management** — (covered: JWKS, JOSE, TLS)
 - covered in your §5.2 sense: existing IETF machinery the use case constrains — exactly-one-current-key rule, mandatory alg; DNSid issues no credentials
- Omitted as incidental: discovery, authentication, authorization, provenance — the draft's explicit non-goals (§3.2, §9.2, §11)

The profile in your notation

```
Durable Accountable Identity Taxonomy Profile
|
|-- Security and Trust
|   |-- Accountability (gap)
|
|-- Identity
|   |-- Agent Naming and Addressing [primary focus; gap]
|   |-- Credential Management (covered: JWKS, JOSE, TLS)
|
|-- Operations and Management
|   |-- Lifecycle Management (gap)
```

- Drop-in §5.3 text exists — framing paragraph, prose bullets, this figure, closing note, mirroring §5.2’s structure, with section citations into -01 throughout

The one divergence, and the offer

- My profile differs from Figure 2 in one place: **Accountability**
 - Figure 2: **(covered: existing audit log conventions)** — correct for DAWN's use case: audit logging of discovery queries inside an operator's own infrastructure
 - This use case: attribution across organizations, after decommissioning, key rotation, domain transfer — there the §4.2.3 identity dependency is the limiting factor
 - §5.1 makes profiles per-use-case, so both markings can be right. I may be wrong about mine.

The offer

- the §5.3 text is drafted and yours if useful, to adopt, edit, or discard
- the draft is on the datatracker — I'd welcome reviewers